

TripTools

Smart Mountain Travel & Expense Companion

A comprehensive, beginner-friendly manual explaining every feature, tool, and calculation formula in simple English with visual workflow diagrams. Optimized for Munnar, South India Highways, and Mountain Road Trips.

Core Modules Covered in This Manual:

1. Ø>Ýí Route Distance Suite (Google Maps Verified 160 KM & 250+ TN Citi
2. &ý Smart Mountain Fuel & Vehicle Rental Cost Calculator
3. Ø=Ü° AI Tour Cost & Multi-Day Budget Predictor
4. Ø=ÜÒ Live Group Expense Ledger & Instant "Who Owes Whom" Splitter
5. Ø=Ü÷ AI Smart Receipt & Bill OCR Scanner
6. Ø=ÜÊ Real-Time Spending Analytics & Over-Budget Gauges
7. Ø=ÜÄ One-Tap PDF Settlement Statement & WhatsApp Trip Sharing

DEVELOPER & PRODUCT DETAILS

Creator / Developer: Bharathkumar E

Live Website: <https://munnartools.vercel.app>

Portfolio: <https://apexassure.vercel.app/>

Support WhatsApp / Phone: +91 8220802736

Email Contact: bharathkumarelango02@gmail.com

Table of Contents

This manual provides clear explanations and step-by-step illustrations for all modules.

Module #	Module Name	Primary Purpose	Page
Module 1	Home & Munnar Sightseeing Guide	Weather, Ghat Helplines & Spot Directory	Page 3
Module 2	Smart Mountain Fuel Calculator	Uphill physics, Vehicle mileage & Fuel Split	Page 4
Module 3	Route Distance & Highway Suite	100% Free 250+ TN cities & Google Verified Distances	Page 5
Module 4	AI Tour Cost & Budget Predictor	Predict total tour expenses for groups in seconds	Page 6
Module 5	Live Group Expense Ledger & OCR	Record bills, AI scanner & "Who Owes Whom"	Page 7
Module 6	Spending Analytics & PDF Generator	Visual charts, PDF statements & WhatsApp sharing	Page 8
Module 7	Offline Mountain Mode & Pro Tips	Using tools in zero-network ghats & hill safety	Page 9

End-to-End Travel Planning Flowchart

STEP 1

Find Exact Route & Distance
Select your city (e.g. Madurai " Munnar = 160 KM). Check 24/7 bunks.

STEP 2

Calculate Mountain Petrol Split
Choose vehicle (Car/Bike/EV). Apply ghat uphill penalty and split per head.

STEP 3

Predict Total Tour Budget
Select duration & luxury level. Generate daily stay, food & sightseeing costs.

STEP 4

Record Live Expenses Offline
Log bills or snap receipts with AI OCR. Real-time equal or custom bill split.

STEP 5

Generate PDF Settlement Sheet
Download official trip statement and share debt settlement on WhatsApp.

Module 1: Home Dashboard & Sightseeing Guide

The Home Dashboard provides immediate situational awareness upon opening the app. It equips travelers with live weather conditions, emergency contacts, and tourist spots.

Feature Name	How to Use It	Why It Is Useful for You
Live Mountain Weather & Mist Forecast	Displayed at the top of the Home tab. Shows temperature, humidity, and mist status.	Helps you choose the right time to start ghat drives before heavy 5 PM fog sets in.
24/7 Emergency Ghat Helplines	Direct tap-to-call buttons for Police (100), Ambulance (108), Forest Dept, and Fire.	Instant emergency calling during vehicle breakdowns or medical alerts in mountain passes.
Munnar Top Attractions Directory	Browse photo cards of Top Station, Eravikulam, Kolukkumalai, Tea Museum, etc.	View entry ticket costs, opening timings, best photography hours, and altitude.
Quick Category Filters	Filter spots by "Waterfalls", "Viewpoints", "Sunrise/Trek", and "Tea Gardens".	Quickly plan a custom daily itinerary based on your group's personal preferences.

+P Top Recommended Attractions & Entry Fees

Attraction Name	Key Highlights	Timings	Approx. Ticket
Eravikulam National Park	Nilgiri Tahr endangered goat, Anamudi View	7:30 AM - 4:00 PM	1200 (Adults)
Kolukkumalai Sunrise	World's Highest Tea Estate & 4x4 Jeep Safari	4:30 AM - 9:00 AM	12,500 / Jeep
Top Station	Panoramic cloud-bed valley view bordering Tamil Nadu	6:00 AM - 6:00 PM	150 / Person
Mattupetty Dam & Lake	Speed boating, elephant sightings, serene lake	9:30 AM - 5:00 PM	1500 / Speedboat
Cheeyappara & Valara Falls	Multi-tier roadside waterfall on Kochi-Munnar NH	Open 24/7	Free Entry

Module 2: Smart Mountain Fuel & Rental Calculator

Mountain driving in low gears (2nd & 3rd) consumes 20% to 35% more fuel than highway cruising. This calculator uses calibrated physics to predict exact liters, petrol costs, and split per friend.

Input Field	Description & How to Set It	Impact on Calculation
Total Distance (KM)	Enter manual KM or tap "Route Finder" to load verified highway distances.	Base multiplier for fuel consumption and rental usage.
Vehicle Type Selector	Choose Hatchback, Sedan, SUV, Bike/Scooter, Royal Enfield 350, or EV.	Auto-fills standard highway mileage and adjusts ghat torque penalties.
Terrain Penalty Mode	Select "Highway Only" (0% penalty) or "Mountain Ghats" (+25% fuel penalty).	Compensates for steep uphill climbing and heavy hairpin braking.
Daily Vehicle Rental Rate	Optional field. Enter self-drive car or rented bike rental per day (e.g. 1500/day).	Combines fuel cost + rental price into a unified single trip total.
Number of Travelers	Enter how many people are sharing the ride (e.g. 4 friends in a car).	Calculates the exact per-person share to collect without arguments.

Real-World Mountain Fuel Formula Explained

1. Effective Mountain Mileage = Standard Mileage ÷ (1 + Ghat Penalty Percentage)

2. Liters Needed = Total Distance (KM) ÷ Effective Mountain Mileage

3. Total Fuel Cost = Liters Needed × Fuel Price per Liter (¹)

4. Cost Per Person = (Total Fuel Cost + Total Rental Fee) ÷ Passenger Count

Module 3: Route Distance & Highway Intelligence Suite

A 100% Free, zero-API-cost navigation intelligence suite covering all 38 Tamil Nadu districts and major South Indian corridors calibrated exactly against verified Google Maps routes.

Popular Route Corridor	Highway Route	Verified KM	Driving Time
Madurai "" Munnar Town	NH 85 via Theni, Bodi Mettu & Gap Road	160 KM	3h 45m
Coimbatore "" Munnar T	SH 17 via Pollachi & Udumalpet Ghats	158 KM	4h 15m
Kochi / Ernakulam "" Mu	NH 85 via Muvattupuzha, Kothamangalam & Adimali	128 KM	3h 30m
Chennai "" Munnar Town	NH 45 / NH 85 via Trichy, Dindigul & Theni	575 KM	9h 30m
Bangalore "" Munnar To	NH 44 via Salem, Dindigul & Theni	480 KM	8h 45m
Theni "" Munnar Town	NH 85 Direct Mountain Pass	84 KM	2h 15m
Bodi (Bodinayakanur) ""	Lockhart Gap Road Mountain Corridor	68 KM	1h 55m
Tirunelveli "" Munnar To	NH 44 / NH 85 via Virudhunagar & Theni	245 KM	5h 15m

&ý Valuable Mountain Highway Road Intelligence

&ý Last 24/7 Fuel Bunk Strategy:

Top up full tank at Theni Bypass or Adimali base bunks. Hill station pumps in Munnar town close at 8:00 PM.

Ø<BÔp Ghat Road & Hairpin Curves:

Routes feature 15-20 hairpin curves. Never ride neutral downhills — always use 2nd/3rd gear engine braking.

& FASTag Tolls & Scenic Stops:

Displays exact toll checkpoints and scenic photo spots (Cheeyappa Falls, Lockhart Gap Viewpoint).

&i 1-Tap Fuel Integration:

Tap "Apply 160 KM to Fuel Calculator" to automatically transfer the distance into the fuel split engine.

Module 4: AI Tour Cost & Budget Predictor

Plan complete tour budgets before leaving home. The AI engine predicts realistic expenses based on group size, trip duration, and chosen travel style (Backpacker, Moderate, Luxury).

Category	Budget / Backpacker	Moderate / Family	Luxury / Honeymoon
Stay & Hot	Homestays (₹800 - ₹1,500/night)	3-Star Resorts (₹2,500 - ₹4,500)	5-Star Valley Villas (₹7,000+)
Food & Din	Local Kerala Mess (₹1750/day)	Multi-Cuisine Cafes (₹1750/day)	Fine Dining Buffets (₹1,500/day)
Transport	KSRTC Public Bus & Shared Auto	Private Cab / Self-Drive Car	Dedicated Innova Crysta / SUV
Sightseei Jeep	National Park & Dam entry	Tea Museum & Boating	Kolukkumalai Sunrise 4x4 Jeep Safari
Emergency Buffer	₹1500 / Person Reserve	₹1,500 / Person Reserve	₹3,000 / Person Reserve

How to Use the Predictor in 3 Quick Steps:

1. Set Duration & People: Slide trip length (e.g. 3 Days / 2 Nights) and group size (e.g. 4 Travelers).
2. Select Travel Style: Tap "Budget", "Standard", or "Luxury" to auto-tune daily spending rates.
3. View Grand Total & Split: See the complete estimated budget + per-person share instantly.
4. Set as Target Budget: Tap "Apply as Trip Budget" to monitor actual spending against this goal!

Module 5: Live Group Expense Ledger & OCR Scanner

Eliminate money disputes during group vacations. Record bills offline, split costs fairly, and calculate exact net balances ("Who owes whom") with 1-tap AI bill scanning.

Feature	Description	How It Helps Your Group
1 - Tap Add Expense	Tap the floating (+) button to log amount, category, payer, and split participants.	Records expenses in under 5 seconds with offline local memory.
AI Smart Receipt Scanner	Snap a photo or upload hotel bills and food receipts directly from mobile.	Auto-extracts bill total, date, and category tags without manual typing.
Smart Bill Splitter	Choose "Equal Split" across all members or "Custom Split" for selective diners.	Prevents unfair splitting when only some friends order non-veg or specialty dishes.
"Who Owes Whom" Engine	Algorithm minimizes transaction count to settle all group debts in fewest payments.	Shows simple clear settlements: e.g., "Praveen pays ₹450 to Bharath".

Example Group Debt Settlement Calculation

Friend Name	Total Paid by Person	Fair Share Owed	Net Settlement Balance
Bharath (Payer 1)	₹4,800 (Resort Bookings)	₹2,000	+₹2,800 (Will Receive)
Karthik (Payer 2)	₹2,200 (Fuel & Jetty)	₹2,000	+₹200 (Will Receive)
Dinesh (Member)	₹1,000 (Dinner)	₹2,000	-₹1,000 (Owes Bharath)
Praveen (Member)	₹0 (Joined trip)	₹2,000	-₹2,000 (Owes ₹1,800 to Bharath, ₹200 to Karthik)

Module 6: Spending Analytics & PDF Report Generator

Keep your spending disciplined and export formal travel statements at the end of the trip. Generate official audit PDF reports or share formatted text summaries on WhatsApp.

Feature Name	Functionality	Output & Benefit
Budget Gauge	Visual meter displaying % of budget used (Green < 75%, Amber 75-95%, Red > 95%).	Gives instant visual warning before your group accidentally exceeds the trip budget.
Category Breakdown Chart	Interactive distribution of Food, Stay, Fuel, Activities, and Shopping.	Identifies where the majority of vacation money is being spent.
Formal PDF Report Generator	1-Tap compiles complete expense tables, payer ledger, and debt settlement sheet.	Downloadable print-ready PDF statement for personal records or office reimbursement.
WhatsApp Instant Share	Formats the complete trip settlement summary into a clean WhatsApp message.	Post directly into your friends' trip group chat with one tap.

Example WhatsApp Trip Statement Output

Munnar Vacation Settlement Statement

***Total Trip Spent: ₹8,000 (4 Travelers • ₹2,000 / Person)**

Stay: ₹4,800 | Fuel: ₹2,200 | Food: ₹1,000

Settlement Summary:

- Dinesh pays ₹1,000 to Bharath
- Praveen pays ₹1,800 to Bharath & ₹200 to Karthik

Generated via TripTools (<https://munnartools.vercel.app>)

Module 7: Offline Mountain Mode & Safety Guidelines

Munnar's high-altitude valleys and ghat passes (Gap Road, Top Station, Marayoor) often have zero cellular connectivity. Here is how TripTools ensures uninterrupted offline functionality.

Offline Feature	How It Works With 0 Mobile Signal	Reliability Guarantee
Service Worker A	Stores full HTML/CSS/JS bundle on device storage.	Loads in < 0.1 seconds with zero internet.
Local Expense P	All bills & group splits save directly into browser storage.	Data is never lost when closing or refreshing tabs.
250+ Embedded Coordinates	Exact GPS coordinates and distances built into app code.	Search and calculate routes completely offline.
Battery & Data O	Zero continuous background polling or battery drain.	Preserves maximum mobile battery on long drives.

Essential Munnar Mountain Driving Safety Rules

- Uphill Right-of-Way: Vehicles climbing uphill always have the right of way. Give way safely on narrow turns.
- Engine Braking on Downhills: Never coast down mountain slopes in Neutral. Keep vehicle in 2nd or 3rd gear.
- Heavy Evening Fog: Gap Road and Top Station experience dense mist after 5:00 PM. Use low-beam yellow fog lamps.
- Wildlife Safety (Chinnar / Marayoor): Watch for wild elephants crossing between 6:00 PM and 6:00 AM. Never honk.